

Lecture 10: Sports Betting

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10.1 Example: A Card Game With a Real Edge

Before discussing sportsbooks, start with a betting game where the probability can be computed exactly.

10.1.1 Instructions

1. Split into pairs and grab a standard 52-card deck. Let player A be the dealer and player B be the bettor. Each player starts with a bankroll of \$100.
2. Player A keeps the cards face down and flips them over one at a time until 30 cards are revealed.
3. Before each flip, player B guesses whether the next card will be red or black, and bets as much of their current bankroll as they want that their guess is correct **at even odds**.
4. After each flip, record player B 's bankroll.
5. After 30 flips, change roles and repeat the game. Whoever finishes with the larger bankroll wins.

10.1.2 Where Does the Edge Come From?

Suppose t cards have been revealed. Let R_t be the number of red cards already revealed and K_t be the number of black cards already revealed, so $R_t + K_t = t$. Then the probability that the next card is red is

$$p_{\text{red},t} = \frac{26 - R_t}{52 - t},$$

and the probability that the next card is black is

$$p_{\text{black},t} = \frac{26 - K_t}{52 - t}.$$

The color choice is straightforward: bet on the color with the larger remaining count. If more black cards than red cards have already appeared, there are more red cards left in the deck, so the bettor should guess red.

Example 10.1 (A Small Card-Counting Edge). Suppose 10 cards have been revealed: 7 black and 3 red. Then there are

23 red cards and 19 black cards

remaining in the 42-card deck. The probability that the next card is red is

$$p_{\text{red}} = \frac{23}{42} \approx 0.548.$$

At even odds, this is a positive expected-value bet: a \$1 bet wins \$1 with probability 0.548 and loses \$1 with probability 0.452. The expected profit is

$$0.548(1) + 0.452(-1) = 0.096.$$

So the bettor should bet red. But how much should the bettor stake? That is the main question of this lecture.

This card game has two useful features. First, the bettor can calculate the true probability exactly. Second, the bettor still has to decide how much to risk. Sports betting has the same two questions, except the first one is much harder:

- Does the price give us positive expected value?
- If so, how much of our bankroll should we bet?

Sports bettors often call positive expected value an **edge**. In these notes, we will keep the comparison explicit: estimate a probability, compare it to the break-even probability implied by the price, and then use Kelly to size the bet.

10.2 Sportsbook Odds and Break-Even Probabilities

A sportsbook does not directly say, “we think this team has probability p of winning.” Instead, it quotes a price. To study sports betting mathematically, we first need to translate prices into probabilities.

10.2.1 Decimal Odds

Throughout the derivation, we will use **decimal odds**. If a bet has decimal odds d , then a \$1 stake returns d dollars if the bet wins. This includes the original \$1 stake. Therefore the net profit from a winning \$1 bet is

$$d - 1.$$

If the bet loses, the bettor loses the \$1 stake.

Example 10.2 (Decimal Odds). If a bet has decimal odds $d = 2.50$, then a \$1 bet returns \$2.50 if it wins. The net profit is \$1.50. If the bet loses, the bettor loses \$1.

10.2.2 Converting American Odds to Decimal Odds

Most American sportsbooks quote prices using American odds. Positive American odds tell us the profit on a \$100 bet. Negative American odds tell us how much must be staked to win \$100.

If the American odds are $a > 0$, then

$$d = 1 + \frac{a}{100}.$$

If the American odds are $a < 0$, then

$$d = 1 + \frac{100}{|a|}.$$

Example 10.3 (Underdog Odds). Suppose the Sixers are offered at +150. Then

$$d = 1 + \frac{150}{100} = 2.50.$$

A \$1 bet returns \$2.50 if the Sixers win, so the net profit is \$1.50.

Example 10.4 (Favorite Odds). Suppose the Celtics are offered at -150. Then

$$d = 1 + \frac{100}{150} = 1.667.$$

A \$1 bet returns \$1.667 if the Celtics win, so the net profit is \$0.667.

10.2.3 Break-Even Probability

For a \$1 bet at decimal odds d , let q be the win probability that makes the bet exactly fair. The expected profit is

$$q(d - 1) + (1 - q)(-1).$$

Setting this equal to 0 gives

$$\begin{aligned} q(d - 1) - (1 - q) &= 0 \\ qd - 1 &= 0 \\ q &= \frac{1}{d}. \end{aligned}$$

Definition 10.5 (Break-Even Probability). *For decimal odds d , the break-even probability is*

$$q = \frac{1}{d}.$$

A bettor needs to win more often than q for the bet to have positive expected value.

Example 10.6 (Exchange Price as Break-Even Probability). Prediction exchanges often quote a binary contract price $c \in (0, 1)$ instead of odds. One share costs c dollars and pays \$1 if the event happens and \$0 otherwise. Ignoring fees, the profit is

$$1 - c \quad \text{if the event happens,} \quad -c \quad \text{otherwise,}$$

so the break-even probability solves

$$q(1 - c) + (1 - q)(-c) = 0,$$

which gives

$$q = c.$$

So on an exchange, the posted price is already the break-even probability.

In practice, platforms may charge trading fees, so the effective break-even probability can be slightly above the posted price. If the trader pays a fee of f dollars per share when entering the position and then holds to resolution, the profit becomes

$$1 - c - f \quad \text{if the event happens,} \quad -c - f \quad \text{otherwise.}$$

The break-even probability then solves

$$q(1 - c - f) + (1 - q)(-c - f) = 0 \implies q = c + f.$$

For example, on some exchanges the taker fee is price-dependent. If $c = 0.60$ and the fee is $f = 0.0072$ per share, then the total entry cost is 0.6072, the winning net profit is

$$1 - 0.60 - 0.0072 = 0.3928,$$

and the effective decimal odds are

$$d = \frac{1}{0.6072} \approx 1.647.$$

So the fee-adjusted break-even probability is

$$q = 0.6072.$$

Remark: On Polymarket, current taker fees are charged at match time using a price-dependent formula; on Kalshi, current trading fees are also charged at execution using a price-dependent schedule. So for these platforms, it is more accurate to think of fees as raising the all-in purchase price slightly above the posted contract price, rather than as a simple settlement haircut on winnings.

Example 10.7 (The Standard -110 Price). A common price on point spreads and totals is -110. The decimal odds are

$$d = 1 + \frac{100}{110} = \frac{210}{110} \approx 1.909.$$

Therefore the break-even probability is

$$q = \frac{1}{1.909} \approx 0.524.$$

So a bettor laying -110 must win about 52.4% of bets just to break even.

Example 10.8 (The Sixers at +110 (Break-Even)). Suppose the Sixers are offered at +110. The decimal odds are

$$d = 1 + \frac{110}{100} = 2.10.$$

The break-even probability is

$$q = \frac{1}{2.10} \approx 0.476.$$

A bettor who thinks the Sixers win more than 47.6% of the time has a positive expected-value bet. A bettor who thinks the Sixers win less than 47.6% of the time should pass.

10.3 The Sportsbook's Edge: Vig and No-Vig Probabilities

If a sportsbook offered perfectly fair prices on both sides of a two-team game, the two implied probabilities would add to 1. In practice, they usually add to more than 1. The extra amount is the sportsbook's built-in edge, often called the **vig**, **juice**, or **hold**.

Suppose a two-sided market has decimal odds d_A on team A and d_B on team B . The raw implied probabilities are

$$q_A = \frac{1}{d_A}, \quad q_B = \frac{1}{d_B}.$$

The hold is

$$h = q_A + q_B - 1.$$

A simple way to remove the vig is to normalize the two implied probabilities:

$$\tilde{q}_A = \frac{q_A}{q_A + q_B}, \quad \tilde{q}_B = \frac{q_B}{q_A + q_B}.$$

These are called **no-vig probabilities**. They are not guaranteed to be the truth, but they are a useful estimate of what the market is saying after removing the sportsbook margin.

Example 10.9 (A -110/-110 Spread Market). Suppose both sides of a point spread are priced at -110. Then both sides have decimal odds

$$d_A = d_B = 1.909,$$

so the raw implied probabilities are

$$q_A = q_B = \frac{1}{1.909} \approx 0.524.$$

Their sum is

$$q_A + q_B \approx 1.048.$$

The sportsbook is charging about a 4.8% hold. After removing the vig,

$$\tilde{q}_A = \tilde{q}_B = \frac{0.524}{1.048} = 0.500.$$

So the market is basically saying the two sides are equally likely after accounting for the sportsbook's margin.

Example 10.10 (An Uneven Moneyline Market). Suppose a sportsbook lists

$$\text{Celtics} - 150, \quad \text{Knicks} + 130.$$

The decimal odds are

$$d_C = 1 + \frac{100}{150} = 1.667, \quad d_K = 1 + \frac{130}{100} = 2.30.$$

The raw implied probabilities are

$$q_C = \frac{1}{1.667} \approx 0.600, \quad q_K = \frac{1}{2.30} \approx 0.435.$$

Their sum is 1.035, so the hold is about 3.5%. The no-vig probabilities are

$$\tilde{q}_C = \frac{0.600}{1.035} \approx 0.580, \quad \tilde{q}_K = \frac{0.435}{1.035} \approx 0.420.$$

This says that, after removing the sportsbook margin, the market is roughly pricing the Celtics as a 58% favorite and the Knicks as a 42% underdog.

The important lesson is that a bettor should not ask only, "Who will win?" The better question is:

Does my estimated probability beat the break-even probability implied by the price?

10.4 Expected Value: Should We Bet?

Now suppose we are considering one bet. Let

- p be the true probability that the bet wins,
- $\hat{p}$ be our estimate of p ,
- d be the decimal odds offered by the sportsbook,
- X be an indicator for whether the bet wins.

Thus $X \sim \text{Bernoulli}(p)$, where $X = 1$ means the bet wins and $X = 0$ means the bet loses. In the derivation below, p denotes the true win probability. In practice we do not know p , so after deriving the rule we will replace it with an estimate $\hat{p}$.

The profit from a \$1 bet is

$$dX - 1.$$

If $X = 1$, the profit is $d - 1$. If $X = 0$, the profit is -1 . Therefore the expected value of a \$1 bet is

$$\begin{aligned} EV &= \mathbb{E}[dX - 1] \\ &= d\mathbb{E}[X] - 1 \\ &= dp - 1. \end{aligned}$$

The bet has positive expected value if and only if

$$dp - 1 > 0,$$

or equivalently

$$p > \frac{1}{d}.$$

Definition 10.11 (Positive Expected Value Bet). *A bet at decimal odds d is a positive expected value bet if*

$$p > \frac{1}{d}.$$

Since p is unknown in sports betting, a bettor uses an estimate $\hat{p}$ and bets only when $\hat{p}$ is sufficiently above the break-even probability.

Example 10.12 (The Sixers at +110 (EV)). Suppose the Sixers are offered at +110, so $d = 2.10$. The break-even probability is 47.6%.

If our model estimates

$$\hat{p} = 0.55,$$

then the estimated expected profit per \$1 bet is

$$\widehat{EV} = 2.10(0.55) - 1 = 0.155.$$

This means the bettor estimates an average profit of 15.5 cents per \$1 staked. This is a strong positive expected-value bet if the model is accurate.

Example 10.13 (A Favorite Can Still Be a Bad Bet). Suppose a team has a 60% chance to win, but the sportsbook offers it at -200. The decimal odds are

$$d = 1 + \frac{100}{200} = 1.50.$$

The break-even probability is

$$\frac{1}{1.50} = 0.667.$$

Even though the team is more likely to win than lose, the bet has negative expected value:

$$EV = 1.50(0.60) - 1 = -0.10.$$

Being likely to win is not enough. The price has to be good.

Expected value tells us whether a bet is attractive. But it still does not tell us how much to bet. If a bet has positive expected value, should we bet 1% of our bankroll? 10%? Everything?

10.5 Bankroll, Compounding, and the Objective

Suppose our current bankroll is B . If we bet a fraction f of the bankroll on a bet with decimal odds d , then the stake is fB . If the bet wins, our new bankroll is

$$B[1 + f(d - 1)].$$

If the bet loses, our new bankroll is

$$B(1 - f).$$

More compactly, if $X \sim \text{Bernoulli}(p)$ indicates whether the bet wins, then

$$B_1 = B[1 + f(dX - 1)].$$

For a sequence of N bets, with odds d_i , win indicators X_i , and betting fractions f_i , the bankroll evolves as

$$B_N = B_0 \prod_{i=1}^N [1 + f_i(d_i X_i - 1)].$$

This product is the key reason betting is different from a one-shot expected-value calculation. Wealth compounds multiplicatively.

10.5.1 Why Not Maximize Expected Bankroll?

A tempting objective is to maximize $\mathbb{E}[B_N]$. The problem is that this objective can favor extremely aggressive bets. For a single positive expected-value bet, maximizing expected final bankroll over $0 \leq f \leq 1$ often pushes us toward betting as much as possible, even though one loss can destroy the bankroll. Rare huge wins can dominate the average even when there is a meaningful chance of ruin.

Kelly betting uses a different objective: maximize expected logarithmic bankroll.

Definition 10.14 (Log-Growth Objective). *For repeated betting, the Kelly criterion chooses betting fractions to maximize*

$$\mathbb{E}[\log(B_N)].$$

Equivalently, since

$$\log(B_N) = \log(B_0) + \sum_{i=1}^N \log[1 + f_i(d_i X_i - 1)],$$

Kelly maximizes the expected long-run geometric growth rate of the bankroll.

This is not the same as maximizing $\mathbb{E}[B_N]$. The logarithm is useful because it turns multiplicative bankroll growth into additive contributions and heavily penalizes outcomes that move the bankroll close to zero. Intuitively, log-growth values steady compounding and survival: doubling the bankroll helps, but getting wiped out ends the compounding process entirely.

Example 10.15 (Why Betting Everything Is Dangerous). Return to the Sixers at +110 with estimated win probability $\hat{p} = 0.55$. The bet has positive estimated expected value. If a bettor with a \$100 bankroll bets everything, then the bankroll becomes

$$\$210 \quad \text{if the Sixers win,}$$

and

$$\$0 \quad \text{if the Sixers lose.}$$

The expected profit looks attractive, but the bettor has a 45% chance of immediate ruin. A useful staking rule must account for this downside.

10.6 The Kelly Criterion: How Much Should We Bet?

Consider a single bet with decimal odds d and true win probability p . If we bet fraction f of our bankroll, the expected log-growth from this bet is

$$g(f) = p \log[1 + f(d - 1)] + (1 - p) \log(1 - f),$$

where $0 \leq f \leq 1$.

The Kelly fraction is the value of f that maximizes $g(f)$.

10.6.1 Derivation

Differentiate $g(f)$ with respect to f :

$$g'(f) = \frac{p(d - 1)}{1 + f(d - 1)} - \frac{1 - p}{1 - f}.$$

Setting $g'(f) = 0$ gives

$$\frac{p(d - 1)}{1 + f(d - 1)} = \frac{1 - p}{1 - f}.$$

Solving for f ,

$$\begin{aligned} p(d - 1)(1 - f) &= (1 - p)[1 + f(d - 1)] \\ p(d - 1) - pf(d - 1) &= 1 - p + f(1 - p)(d - 1) \\ p(d - 1) - (1 - p) &= f(d - 1)[p + (1 - p)] \\ pd - 1 &= f(d - 1). \end{aligned}$$

Therefore

$$f^* = \frac{pd - 1}{d - 1}.$$

If $pd - 1 \leq 0$, the bet has nonpositive expected value, and the Kelly rule says not to bet. Thus the final one-bet Kelly formula is

$$f^* = \max\left(0, \frac{pd - 1}{d - 1}\right).$$

Definition 10.16 (Kelly Fraction). *For a binary bet with decimal odds d and win probability p , the Kelly fraction is*

$$f^* = \max\left(0, \frac{pd - 1}{d - 1}\right).$$

It is the fraction of current bankroll that maximizes expected log-growth under the model.

10.6.2 Back to the Card Game

At even odds, $d = 2$. The Kelly formula becomes

$$f^* = \max(0, 2p - 1).$$

So in the card game, the optimal stake grows as the deck becomes more imbalanced.

Example 10.17 (Card Game Stake Size). Earlier, after 10 revealed cards, there were 23 red cards and 19 black cards left. The probability of red was

$$p = \frac{23}{42} \approx 0.548.$$

At even odds, the Kelly fraction is

$$f^* = 2(0.548) - 1 = 0.096.$$

So full Kelly would bet about 9.6% of the current bankroll on red. If the bankroll is \$100, the full-Kelly stake is about \$9.60.

Example 10.18 (A Bigger Card-Counting Edge). Suppose 20 cards have been revealed: 15 black and 5 red. Then there are 21 red cards and 11 black cards left, so

$$p_{\text{red}} = \frac{21}{32} = 0.656.$$

At even odds,

$$f^* = 2(0.656) - 1 = 0.312.$$

Full Kelly would bet about 31.2% of the bankroll on red. The larger the probability advantage over break-even, the larger the optimal stake.

10.6.3 Sports Betting Examples

Example 10.19 (The Standard -110 Bet). Suppose a spread bet is priced at -110, so $d = 1.909$. If our model says the bet wins with probability $p = 0.55$, then the Kelly fraction is

$$\begin{aligned} f^* &= \frac{pd - 1}{d - 1} \\ &= \frac{0.55(1.909) - 1}{1.909 - 1} \\ &\approx 0.055. \end{aligned}$$

Full Kelly would bet about 5.5% of bankroll. With a \$1,000 bankroll, this is a \$55 stake.

Example 10.20 (The Sixers at +110 (Kelly)). Suppose the Sixers are priced at +110, so $d = 2.10$, and our model estimates $p = 0.55$. Then

$$\begin{aligned} f^* &= \frac{0.55(2.10) - 1}{2.10 - 1} \\ &= \frac{0.155}{1.10} \\ &\approx 0.141. \end{aligned}$$

Full Kelly would bet about 14.1% of bankroll. This is much larger than the -110 example because the price is much better relative to the same 55% win probability.

Example 10.21 (A Likely Winner, But Still No Bet). Suppose a team has win probability $p = 0.60$, but the sportsbook price is -200, so $d = 1.50$. The Kelly numerator is

$$pd - 1 = 0.60(1.50) - 1 = -0.10.$$

Thus

$$f^* = 0.$$

Kelly agrees with expected value: even though the team is likely to win, the price is too expensive.

10.7 Fractional Kelly and Model Uncertainty

The card game is special because the probability can be computed exactly from the remaining deck. In sports betting, the true win probability p is not observed. We estimate it using data, models, injury reports, matchup information, and market information. Let this estimate be $\hat{p}$.

A plug-in Kelly rule would use

$$\hat{f}_{\text{Kelly}} = \max\left(0, \frac{\hat{p}d - 1}{d - 1}\right).$$

The danger is that $\hat{p}$ may be wrong. If our model overstates the true advantage, full Kelly can bet much too aggressively.

10.7.1 Fixed Fractional Kelly

A common practical adjustment is **fractional Kelly**. Instead of betting the full Kelly fraction, we bet a fixed fraction $K \in (0, 1)$ of it:

$$f_K = K \hat{f}_{\text{Kelly}}.$$

Common choices are $K = 1/2$ for half Kelly or $K = 1/4$ for quarter Kelly.

Example 10.22 (Half Kelly on the Sixers). For the Sixers at +110 with $\hat{p} = 0.55$, full Kelly gave

$$\hat{f}_{\text{Kelly}} \approx 0.141.$$

Half Kelly bets

$$f_{1/2} = \frac{1}{2}(0.141) = 0.0705,$$

or about 7.1% of bankroll. Quarter Kelly bets

$$f_{1/4} = \frac{1}{4}(0.141) = 0.0353,$$

or about 3.5% of bankroll.

10.7.2 Fractional Kelly as Shrinkage

Fractional Kelly can be interpreted as shrinking the model probability toward the break-even probability. Let

$$q = \frac{1}{d}$$

denote the break-even probability. Define the shrunken probability

$$\tilde{p}_K = q + K(\hat{p} - q).$$

Then full Kelly using $\tilde{p}_K$ gives the same stake as K -fractional Kelly using $\hat{p}$:

$$\begin{aligned} \frac{\tilde{p}_K d - 1}{d - 1} &= \frac{[q + K(\hat{p} - q)]d - 1}{d - 1} \\ &= \frac{qd - 1 + Kd(\hat{p} - q)}{d - 1} \\ &= \frac{K(d\hat{p} - dq)}{d - 1} \\ &= K \frac{d\hat{p} - 1}{d - 1}, \end{aligned}$$

since $dq = 1$.

Example 10.23 (Shrinkage Interpretation). For the Sixers at +110, $d = 2.10$, so

$$q = \frac{1}{2.10} \approx 0.476.$$

If $\hat{p} = 0.55$ and $K = 1/2$, then

$$\tilde{p}_{1/2} = 0.476 + \frac{1}{2}(0.55 - 0.476) = 0.513.$$

Half Kelly is equivalent to pretending our probability estimate is 51.3% instead of 55.0%. We still bet because 51.3% is above the break-even probability 47.6%, but we bet much less aggressively.

10.7.3 Uncertainty-Dependent Fractional Kelly

If our probability estimate is very precise, then full Kelly may be a reasonable theoretical benchmark. If the estimate is noisy, biased, or based on a small sample, then the Kelly fraction should be shrunk more aggressively.

Let τ^2 represent the uncertainty in the estimate $\hat{p}$, for example

$$\text{Var}(\hat{p}) = \tau^2.$$

Then we can imagine using a fractional Kelly multiplier $K(\tau)$ satisfying

$$\begin{aligned} \lim_{\tau \rightarrow 0} K(\tau) &= 1, \\ \lim_{\tau \rightarrow \infty} K(\tau) &= 0. \end{aligned}$$

When the model is very certain, we stake closer to full Kelly. When the model is very uncertain, we stake closer to zero.

Example 10.24 (A Small Estimated Advantage Should Usually Stay Small). Suppose a model estimates that a +110 bet wins with probability $\hat{p} = 0.49$. The break-even probability is 0.476, so this is technically positive expected value. Full Kelly gives

$$\hat{f}_{\text{Kelly}} = \frac{0.49(2.10) - 1}{1.10} \approx 0.026.$$

Full Kelly would bet 2.6% of bankroll. But if this estimated advantage comes from a noisy model, quarter Kelly would bet only

$$0.25(0.026) = 0.0065,$$

or about 0.65% of bankroll. Small estimated advantages should not be treated the same way as large, stable, well-understood advantages.

10.8 Complications in Real Sports Betting

The clean Kelly formula assumes a binary bet, known probability p , known odds d , and a bankroll that can be freely divided across independent opportunities. Real sports betting is messier.

10.8.1 Pushes

Some spread and total bets can push. If a bet wins with probability p , loses with probability ℓ , and pushes with probability r , then

$$p + \ell + r = 1.$$

A push returns the stake, so it contributes $\log(1) = 0$ to log-growth. The expected log-growth is

$$g(f) = p \log[1 + f(d - 1)] + \ell \log(1 - f).$$

The same logic applies, but the probabilities must now distinguish wins, losses, and pushes.

Example 10.25 (A Spread With Push Probability). Suppose a point-spread bet at -110 wins 52% of the time, loses 44% of the time, and pushes 4% of the time. The expected profit per \$1 is

$$0.52(0.909) + 0.44(-1) + 0.04(0) = 0.033.$$

This is positive expected value, but the Kelly calculation should use the win and loss probabilities, not just a binary win probability that ignores pushes.

10.8.2 Correlation Across Bets

Kelly sizing is simplest when bets are independent. Sports bets are often correlated. A quarterback injury can affect the side, total, team total, player props, and derivative markets. Betting full Kelly on each one separately can overexpose the bankroll to one underlying piece of information.

Example 10.26 (One Injury, Many Bets). Suppose a model likes the Eagles spread, the Eagles team total over, and the opposing quarterback's passing yards under because of the same injury report. These are not three independent opportunities. If the injury information is wrong or already priced in, all three bets can lose together. A practical Kelly bettor should reduce the combined exposure.

10.8.3 Parlays

A parlay combines multiple legs into one bet. If the legs are independent, the win probability is the product of the leg probabilities. If the legs are correlated, multiplying probabilities can be badly wrong.

Example 10.27 (Two-Leg Parlay). Suppose two independent legs each have win probability 0.55. Then the parlay win probability is

$$0.55^2 = 0.3025.$$

If the parlay pays decimal odds $d = 3.60$, the expected value per \$1 is

$$3.60(0.3025) - 1 = 0.089.$$

But this calculation depends completely on the probability estimate. If the legs are not independent, or if the sportsbook has priced the correlation correctly, the positive expected value may disappear.

10.8.4 Limits, Liquidity, and Bankroll Definition

The formula may say to bet 7% of bankroll, but a sportsbook may limit the stake, move the line, or reject the bet. Also, the bankroll should be defined in advance. It should not be confused with net worth or money needed for ordinary life.

Example 10.28 (The Bankroll Matters). A 5% Kelly stake means different things for different bankrolls. With a \$1,000 betting bankroll, it is a \$50 bet. With a \$50,000 bankroll, it is a \$2,500 bet. Kelly scales with bankroll, so the bankroll number must be chosen carefully before applying the formula.

10.9 Summary

The lecture can be summarized in two questions:

1. **Should we bet?** Compare our estimated probability $\hat{p}$ to the break-even probability $1/d$. Equivalently, compute estimated expected value

$$\widehat{EV} = d\hat{p} - 1.$$

Bet only when the price is favorable.

2. **How much should we bet?** If the bet has positive expected value, the full Kelly benchmark is

$$\hat{f}_{\text{Kelly}} = \max\left(0, \frac{d\hat{p} - 1}{d - 1}\right).$$

In practice, because $\hat{p}$ is uncertain, fractional Kelly is usually more defensible:

$$f_K = K\hat{f}_{\text{Kelly}}, \quad 0 < K < 1.$$

The core idea is simple but powerful: **sports betting is not about picking winners; it is about finding prices that are better than your estimated probabilities, then sizing those bets without overexposing the bankroll.**

References

- [JLK] Kelly, J. L., *A New Interpretation of Information Rate*, Bell System Technical Journal, 1956.